# Core-collapse Supernovae in the Dark Energy Survey

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# Introduction

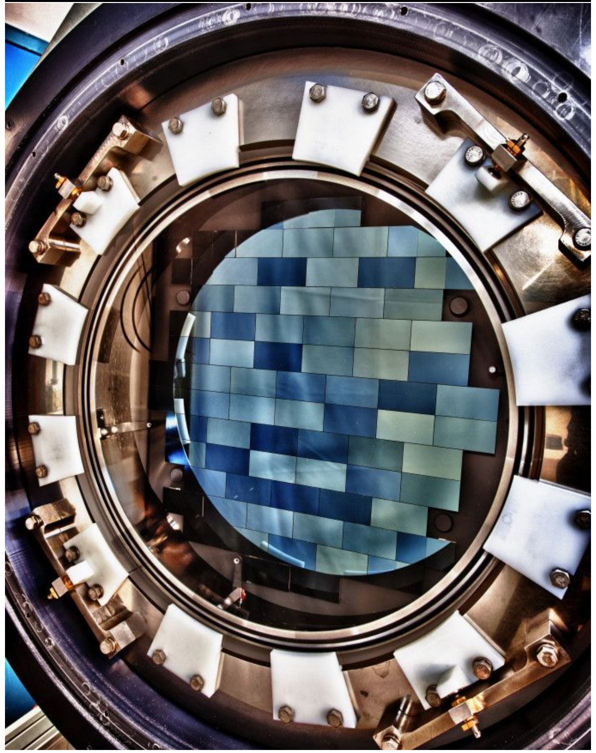
- Core-collapse supernovae (CCSNe): most common type of cosmic explosion, but not well understood
- Their distribution in luminosity - the 'luminosity function' - is the most basic population diagnostic
- This is critical for understanding their presence as a contaminant in SN Ia cosmological samples

# Previous Work

- Two main previous studies looking at CCSNe luminosity functions
- Li et al. 2011
  - Used Lick Observatory Supernova Search, 105 CCSNe
- Richardson et al. 2014
  - Based on full Asiago Supernova Catalogue
  - No uncertainties, for now we have not compared to this

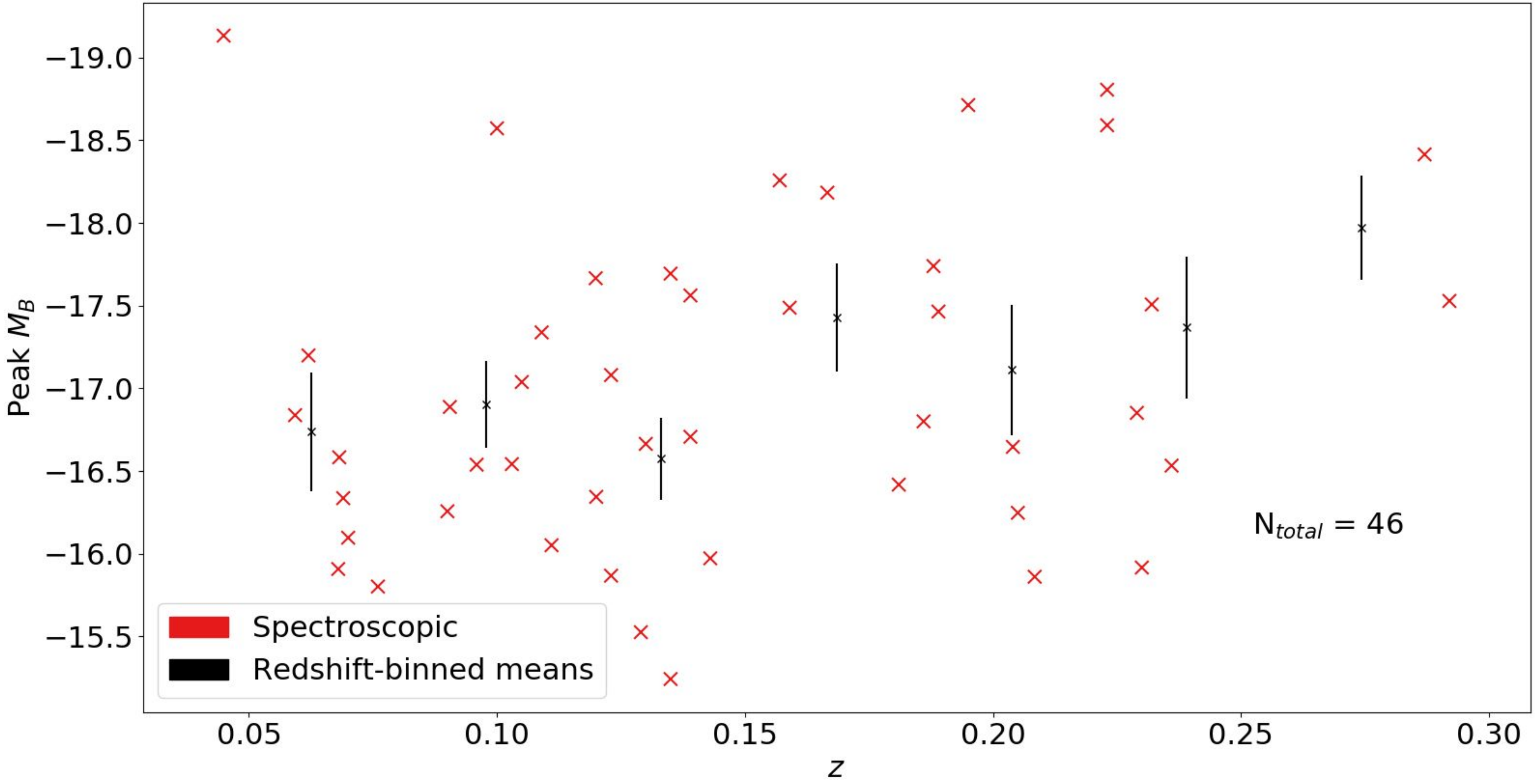
# Dark Energy Survey (DES)

- Aim - “Everything with one survey”, studying dark energy using multiple probes
- DES SN
  - Observed 30 square degrees in four filters every 7 nights for 5 years
  - Very deep survey, ~24 mag
  - Doesn’t pick up many local objects but very good for detecting objects at high redshift

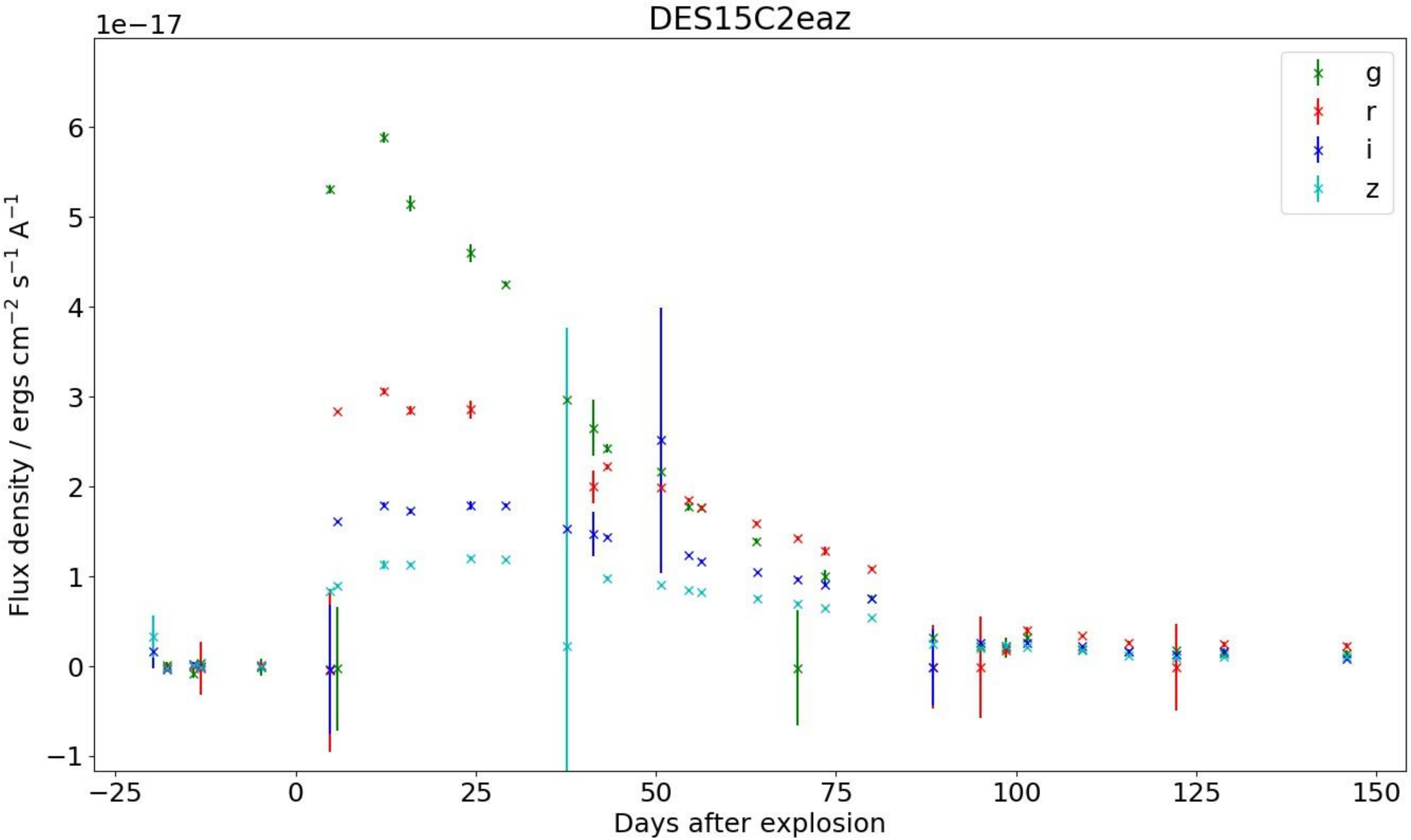


# DES Sample - Spectroscopic

- 70 spectroscopically-confirmed CCSNe in DES, 46 after cuts
  - 37 Hydrogen-rich, 9 stripped-envelope

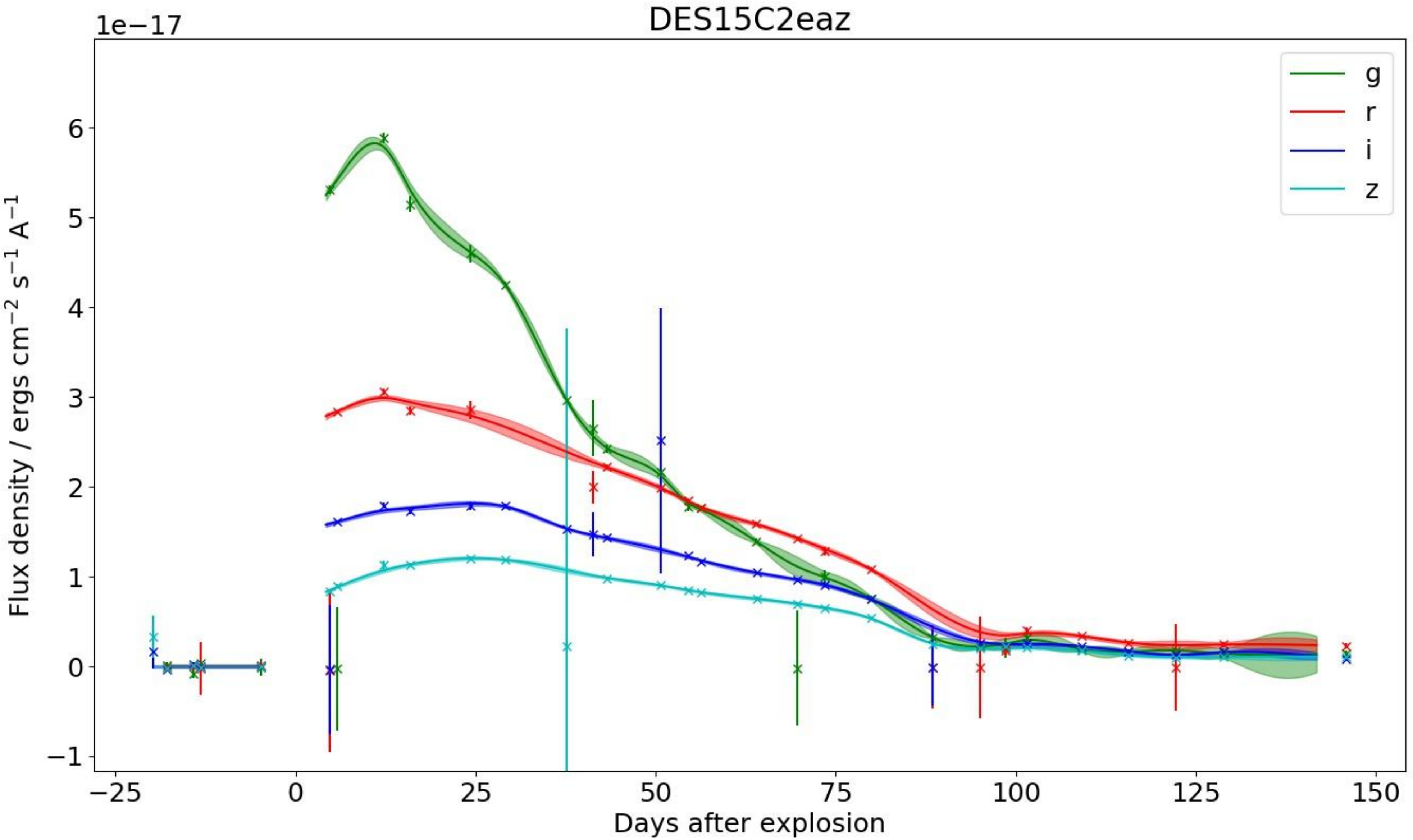


# Analysis



DES15C2eaz
1e−17
Flux density / ergs cm$^{-2}$ s$^{-1}$ A$^{-1}$
Days after explosion
g
r
i
z
−1
0
1
2
3
4
5
6
−25
0
25
50
75
100
125
150

# Analysis



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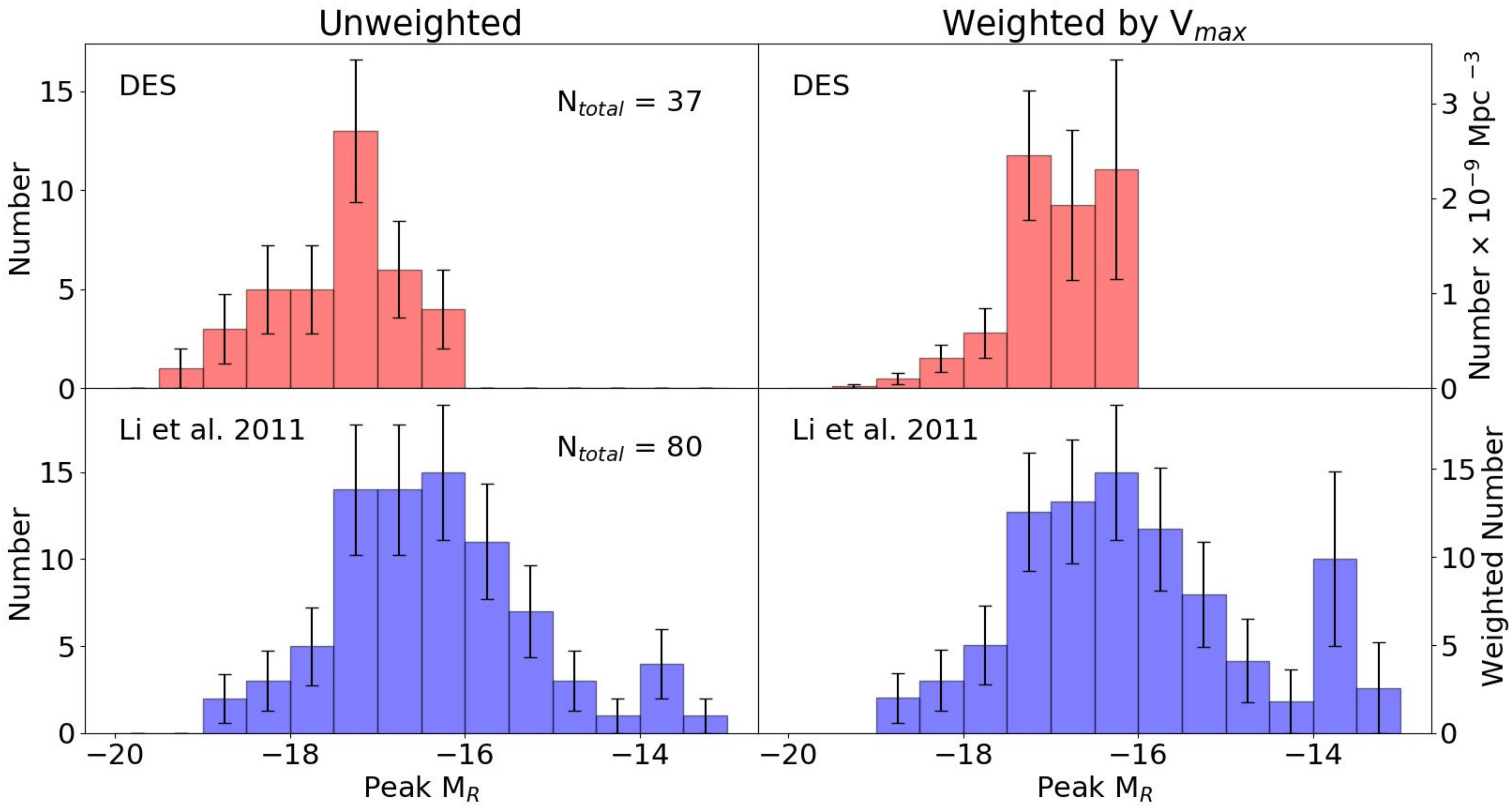
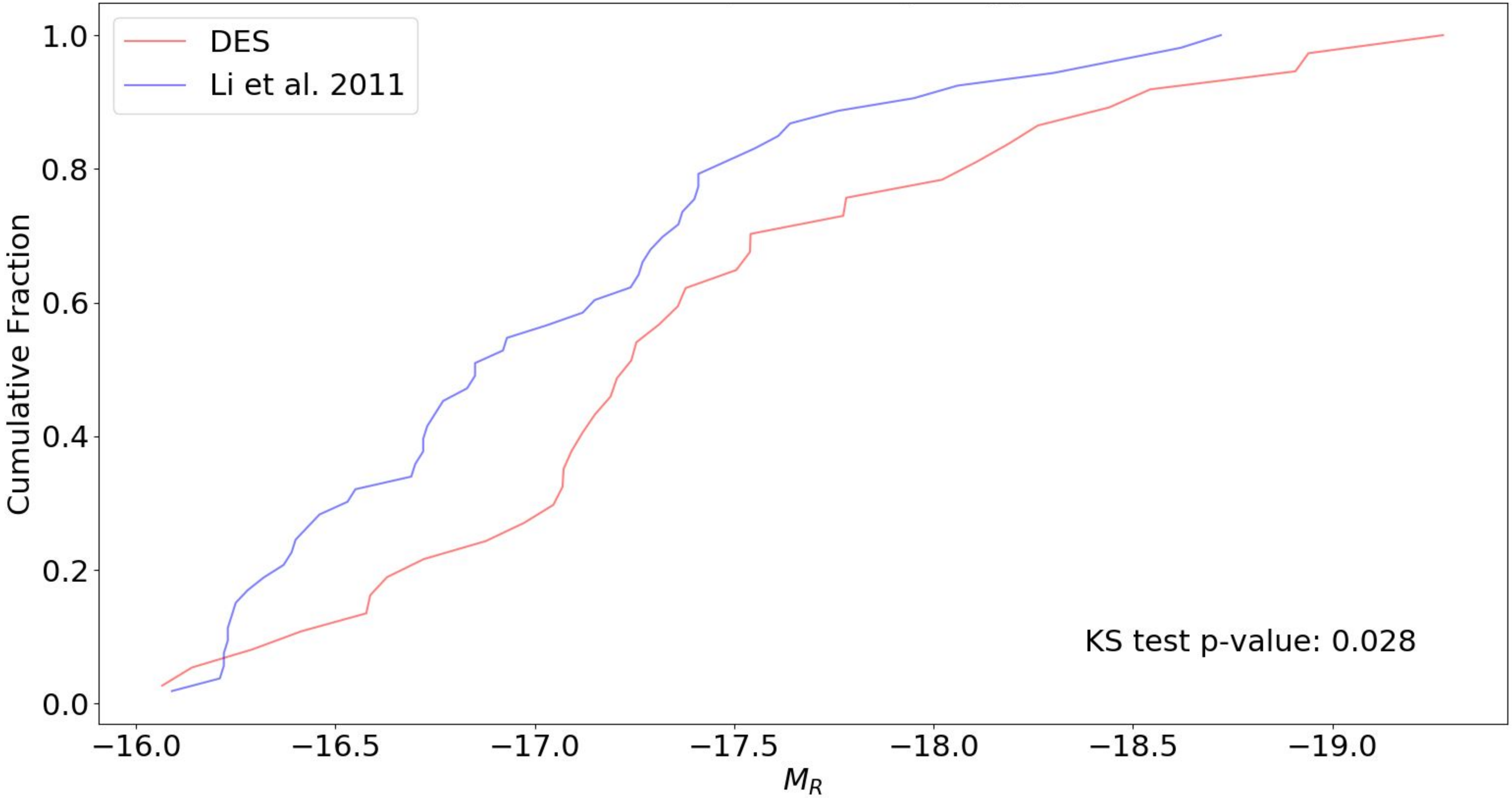
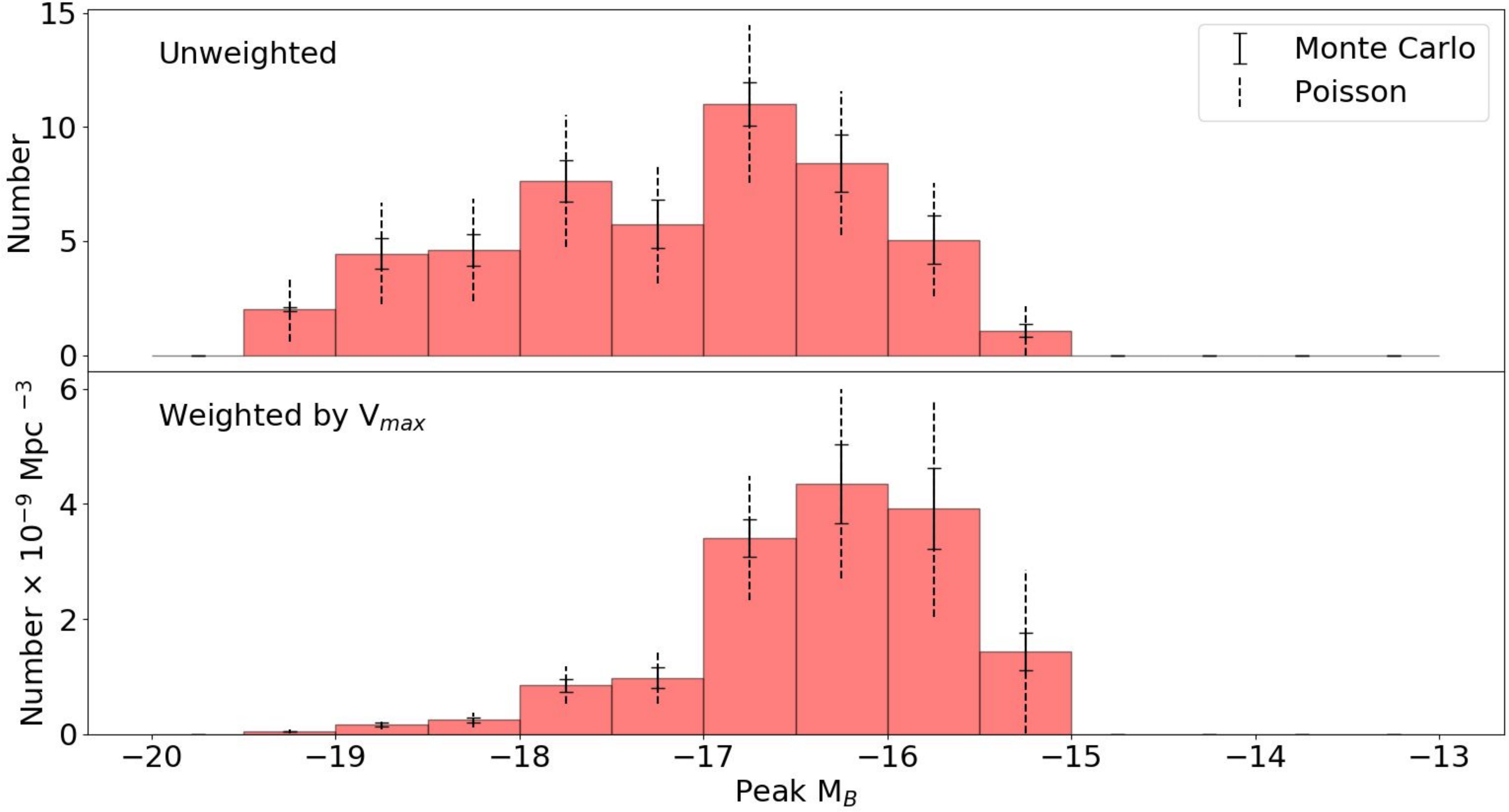
# Results - Luminosity Functions (Spectroscopic SNe II - R band)



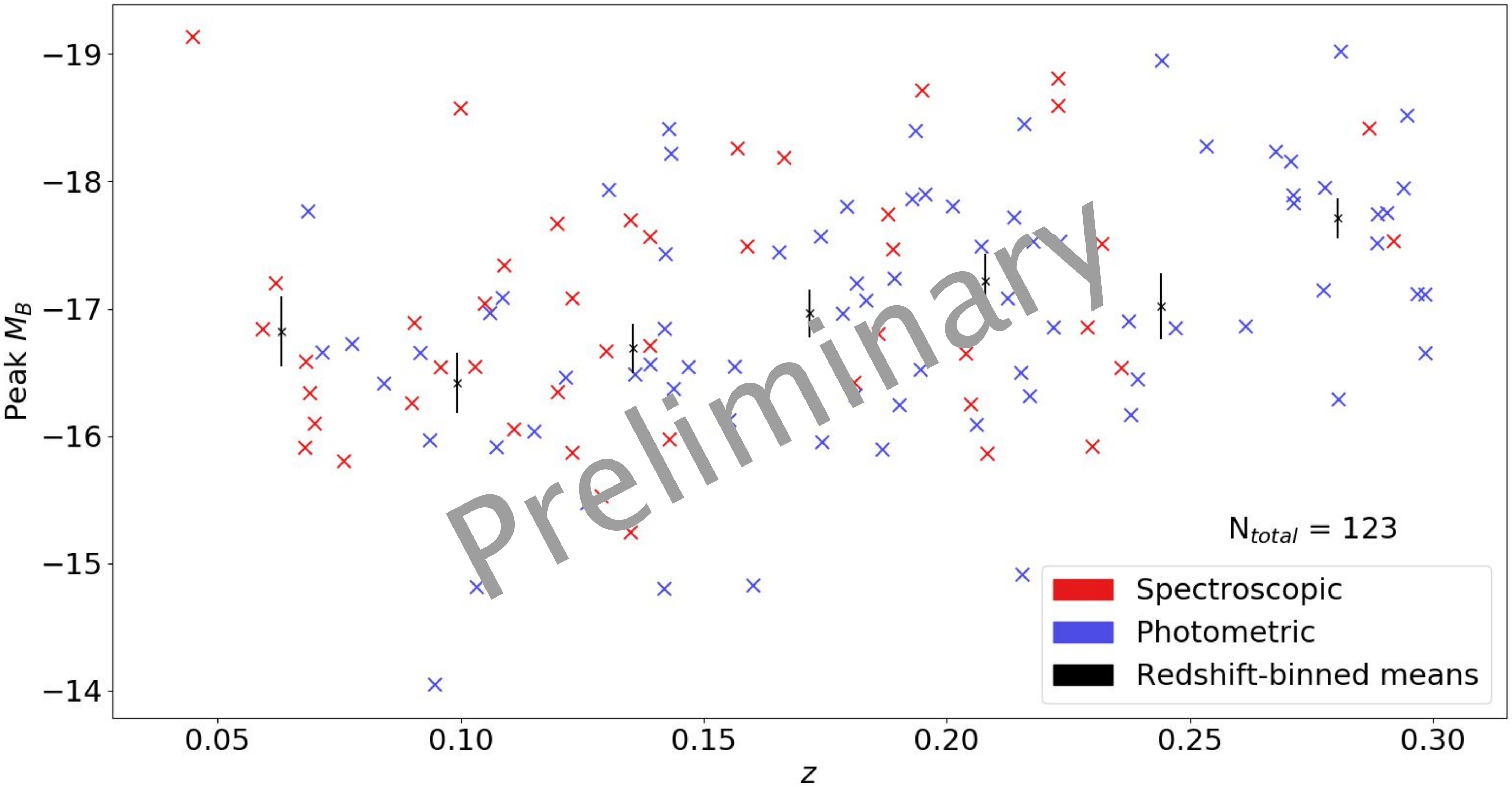
# Results - KS Test (Spectroscopic SNe II - R band)



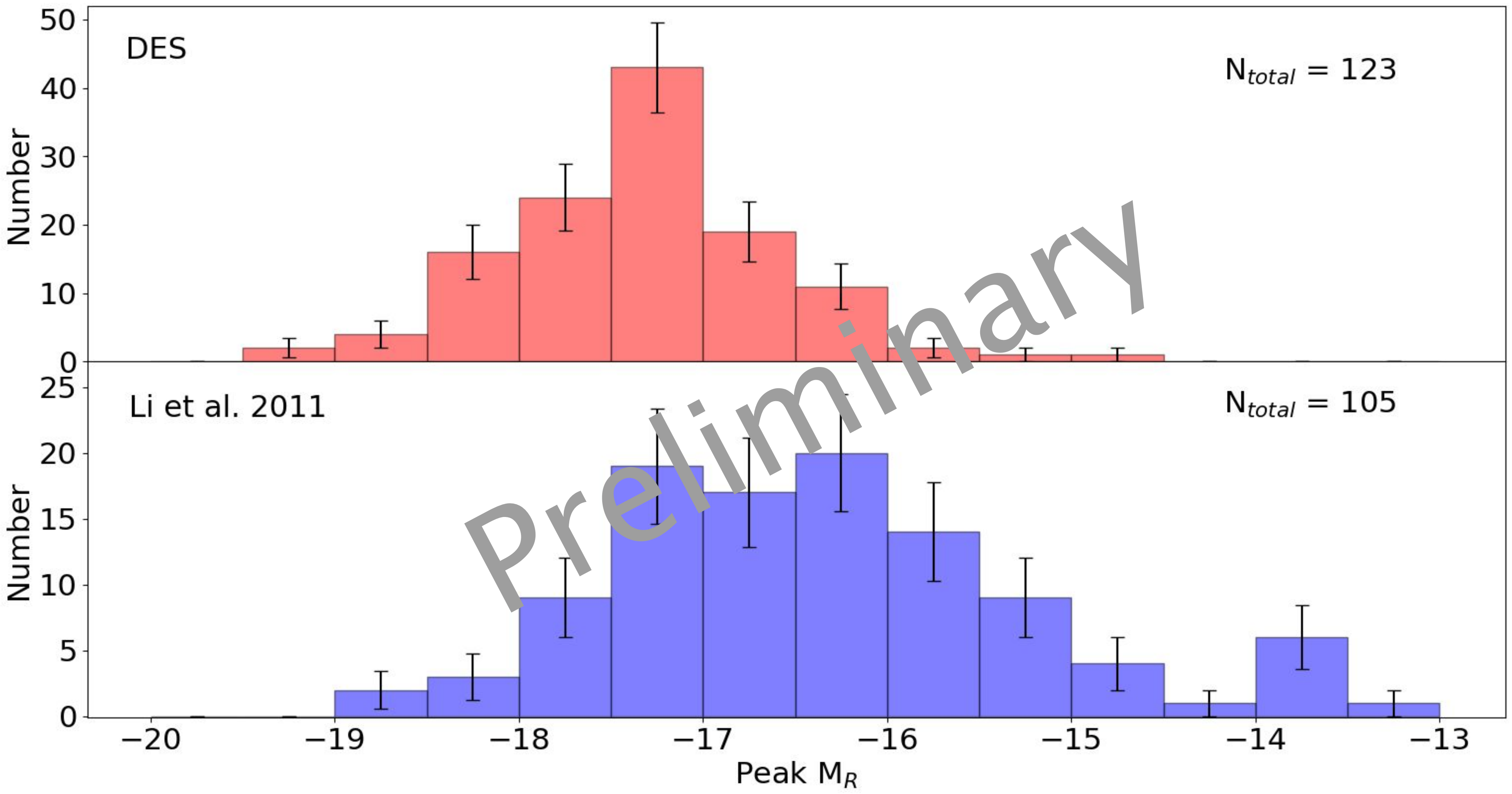
# Results - Luminosity Functions (Spectroscopic SNe II - B band)



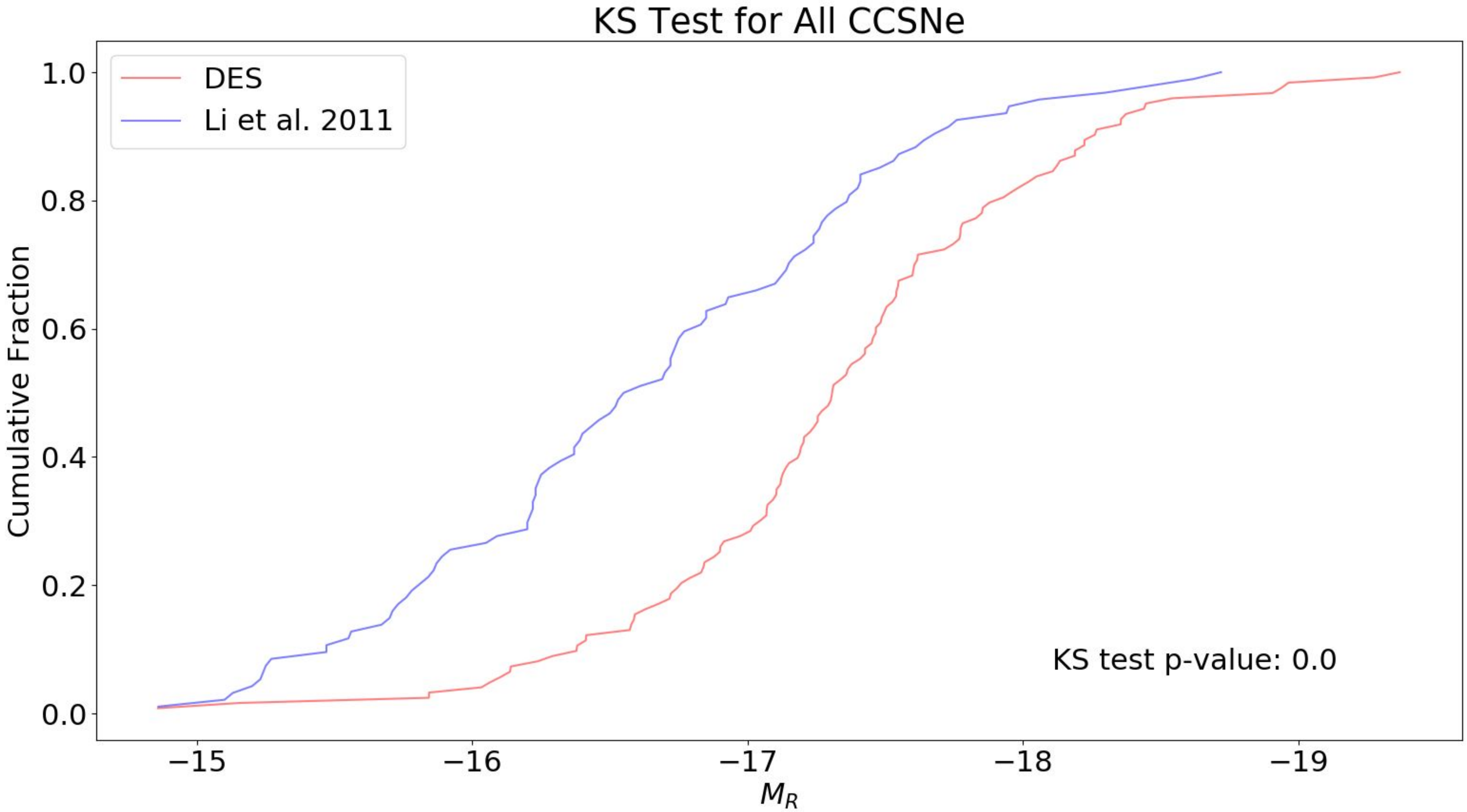
# DES Sample - Spectroscopic and Photometric



# Results - Luminosity Functions (All CCSNe including Photometric)



# Results - KS Test (All CCSNe including Photometric)



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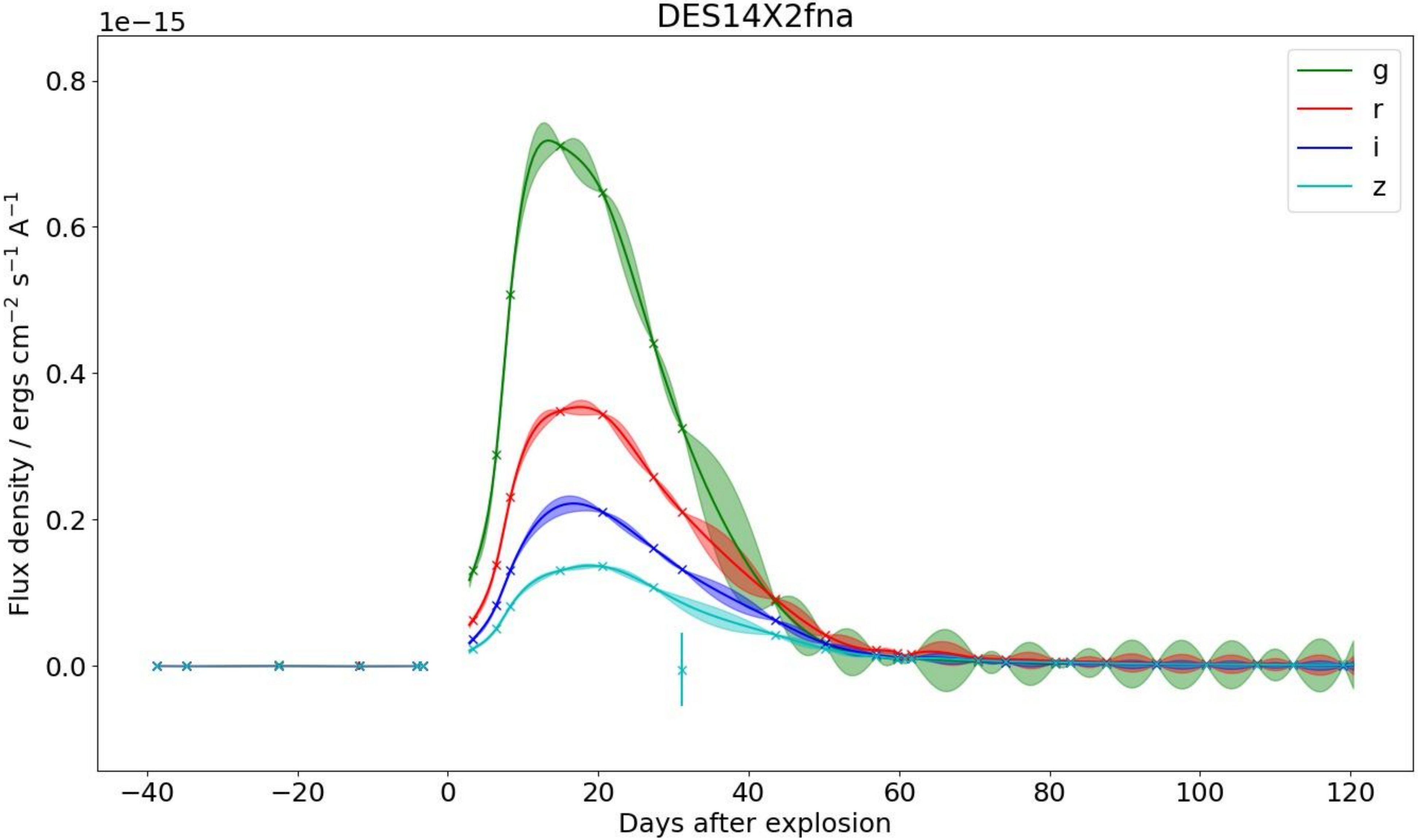
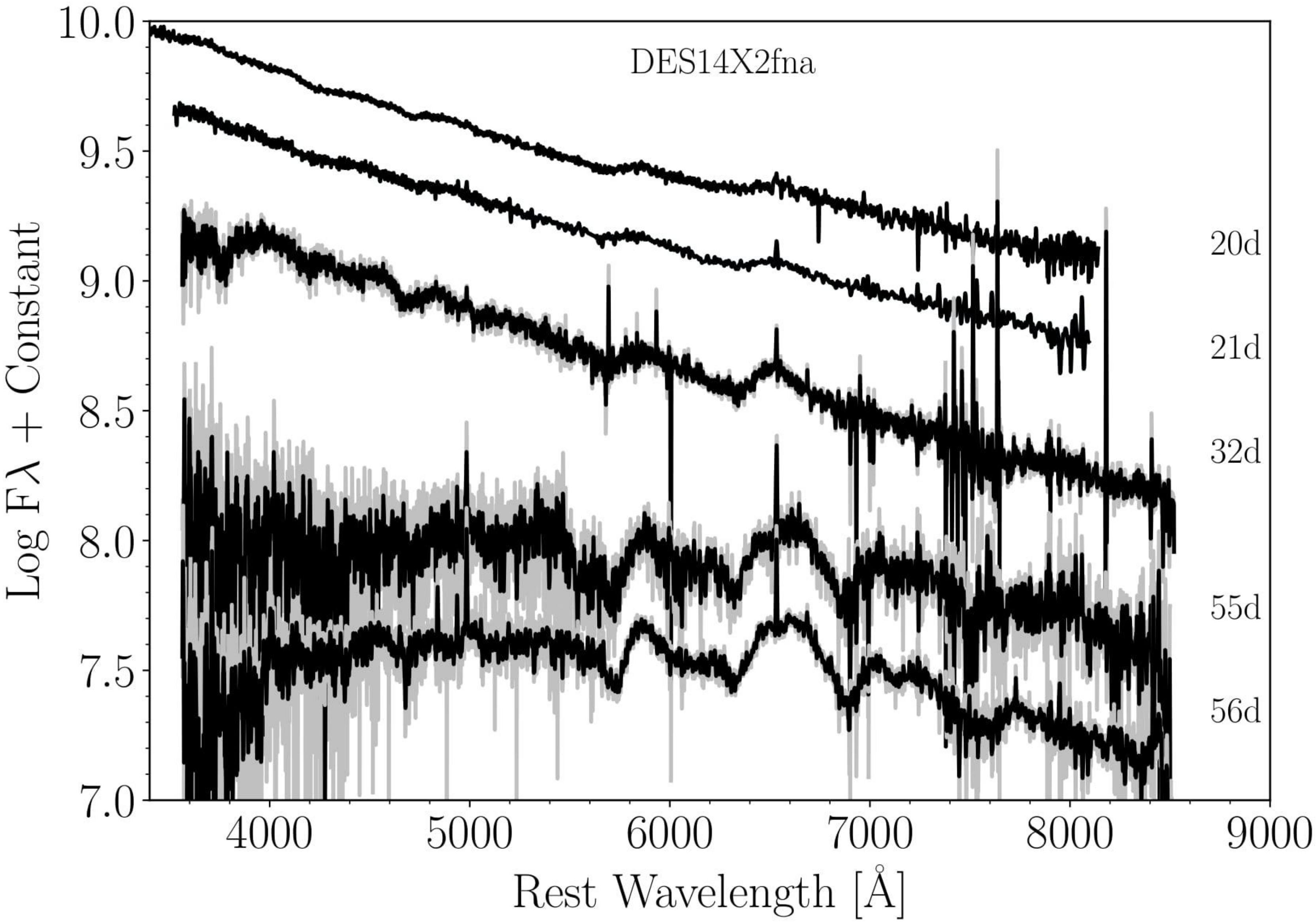
# Summary and next steps

- We have obtained a high-redshift sample of CCSNe from DES consisting of 46 spectroscopic and 77 photometric candidates
- Our analysis has shown significant differences between our sample in DES and the low-redshift sample from Li et al. 2011
- We will follow this up by analysing an intermediate redshift-sample to see if there is an evolution in redshift
- We will refine our photometric sample and separate by subtype to improve our analysis, and examine the effect of uncertainties in our peak luminosities

# Results - DES14X2fna



# Results - DES14X2fna



# Results - DES14X2fna

- Brightest Type IIb SN ever detected, redshifts from both SN spectra and host galaxy
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| Rest-frame filter | Peak absolute magnitude |
|---|---|
| B | -19.22 |
| V | -19.08 |
| R | -19.37 |

- Good example of new objects found in DES
- Need to be aware of these potential very luminous CCSNe when creating Ia samples

# Analysis

- Observed photometry was corrected for Milky Way extinction and interpolated using Gaussian Processes
- Observed-frame light curves were K-corrected to rest-frame BVR light curves
- Peak absolute magnitudes were calculated from rest-frame light curves
- Uncertainties on peak absolute magnitude were estimated using Monte Carlo Methods

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# Results - DES14X2fna